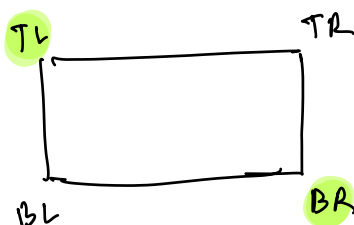
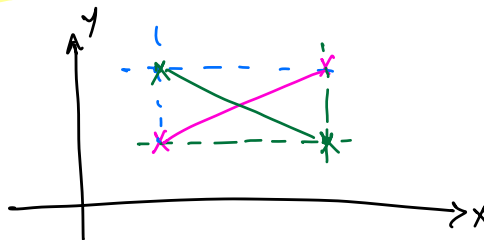


Today's Content

- Submatrix Sum Queries
- Sum of all submatrices
- Maximum Submatrix Sum

Q1 Given a matrix of size $N \times M$. For each query, q , find sum of given sub matrix.
 $\rightarrow$ part of a matrix (rectangle)

	0	1	2	3	4	5
0	7	1	-6	3	12	-2
1	10	5	-2	0	9	7
2	6	7	-3	8	11	3
3	13	-8	-5	12	4	6
4	3	2	1	9	3	9
5	4	3	-2	6	8	8



ex. TL and BR
 $\downarrow$
 $(1, 2)$ $\downarrow$ $(4, 3)$
 sum = 20

			b1	b2		
	0	1	2	3	4	5
0	7	1	-6	3	12	-2
a1 1	10	5	-2	0	9	7
2	6	7	-3	8	11	3
3	13	-8	-5	12	4	6
a2 4	3	2	1	9	3	9
5	4	3	-2	6	8	8

ex. TL and BR
 (a_1, b_1) (a_2, b_2)

Brute force:- Iterate over entire submatrix and get the sum.

sum = 0;

```
for (i = a1; i <= a2; i++) {  
    for (j = b1; j <= b2; j++) {  
        sum += mat[i][j];  
    }  
}
```

$$T.C = O(R * N * M)$$

$$S.C = O(1)$$

$pf[i] =$ sum of elements from $[0, i]$

$pf[i][j] =$ sum of all elements from $[0, 0]$ to $[i, j]$

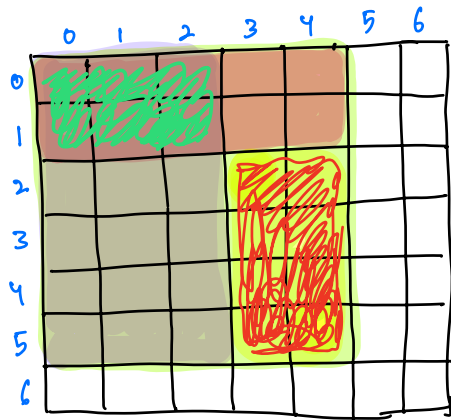
$$pf[3][2] = \text{sum}([0, 0] - [3, 2])$$

$$pf[0][4] = \text{sum}([0, 0] - [0, 4])$$

$$pf[2][4] = \text{sum}([0, 0] - [2, 4])$$

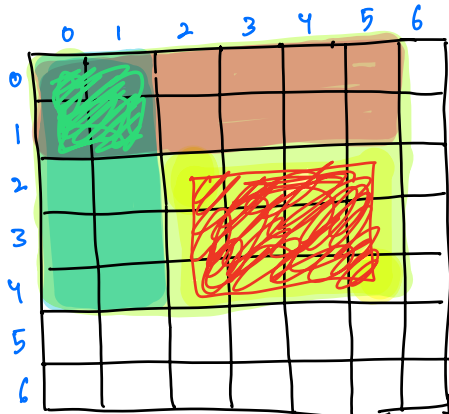
	0	1	2	3	4	5
0	7	1	-6	3	12	-2
1	10	5	-2	0	9	7
2	6	7	-3	8	11	3
3	13	-8	-5	12	4	6
4	3	2	1	9	3	9
5	4	3	-2	6	8	8

$$Q_1:- \overset{TL}{(2,3)} \rightarrow \overset{BR}{(5,4)}$$

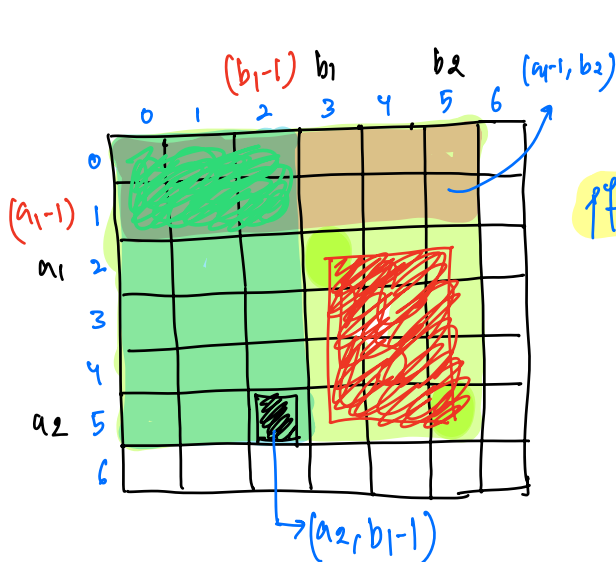


$$pf[5,4] - pf[5,2] - pf[1,4] + pf[1,2]$$

$$Q_2:- \overset{TL}{(2,2)} \rightarrow \overset{BR}{(4,5)}$$



$$pf[4,5] - pf[4,1] - pf[1,5] + pf[1,1]$$



$$Q_3:- \overset{TL}{(a_1, b_1)} \rightarrow \overset{BR}{(a_2, b_2)}$$

$$pf[a_2, b_2] - pf[a_2, b_1-1] - pf[a_1-1, b_2] + pf[a_1-1, b_1-1]$$

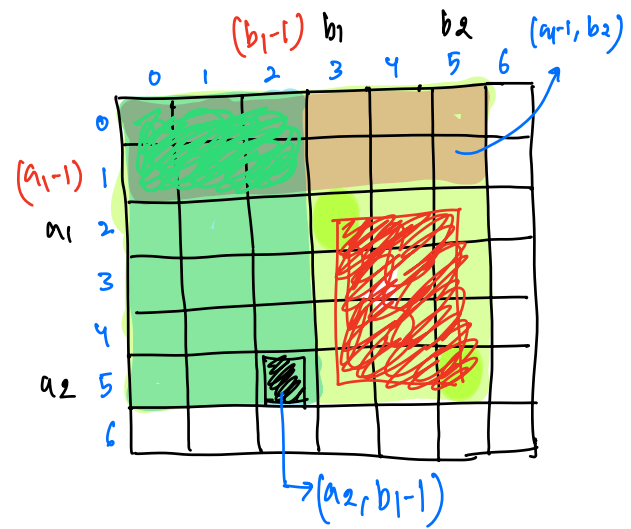
$sum = pf[a_2, b_2]$

if ($b_1 \neq 0$) {
 $sum = sum - pf[a_2, b_1 - 1]$
}

if ($a_1 \neq 0$) {
 $sum = sum - pf[a_1 - 1, b_2]$
}

if ($a_1 \neq 0$ & & $b_1 \neq 0$) {
 $sum = sum + pf[a_1 - 1, b_1 - 1]$
}

return sum;



	0	1	2
0	a_0	b_0	c_0
1	a_1	b_1	c_1
2	a_2	b_2	c_2

Along every row take $pf[i]$

Along every col take $pf[i]$

0	a_0	$a_0 + b_0$	$a_0 + b_0 + c_0$
1	a_1	$a_1 + b_1$	$a_1 + b_1 + c_1$
2	a_2	$a_2 + b_2$	$a_2 + b_2 + c_2$

0	a_0	$a_0 + b_0$	$a_0 + b_0 + c_0$
1	$a_0 + a_1$	$a_0 + b_0 + a_1 + b_1$	$a_0 + b_0 + c_0 + a_1 + b_1 + c_1$
2	$a_0 + a_1 + a_2$	$a_0 + b_0 + a_1 + b_1 + a_2 + b_2$	$a_0 + b_0 + c_0 + a_1 + b_1 + c_1 + a_2 + b_2 + c_2$

Steps:-

1) Construct pf[77] matrix

a) Apply pf on every row

b) Apply pf on every column

2) $sum = pf[a_2, b_2]$

if ($b_1 \neq 0$) {

| $sum = sum - pf[a_2, b_1 - 1]$

}

if ($a_1 \neq 0$) {

| $sum = sum - pf[a_1 - 1, b_2]$

}

if ($a_1 \neq 0$ & & $b_1 \neq 0$) {

| $sum = sum + pf[a_1 - 1, b_1 - 1]$

}

return sum;

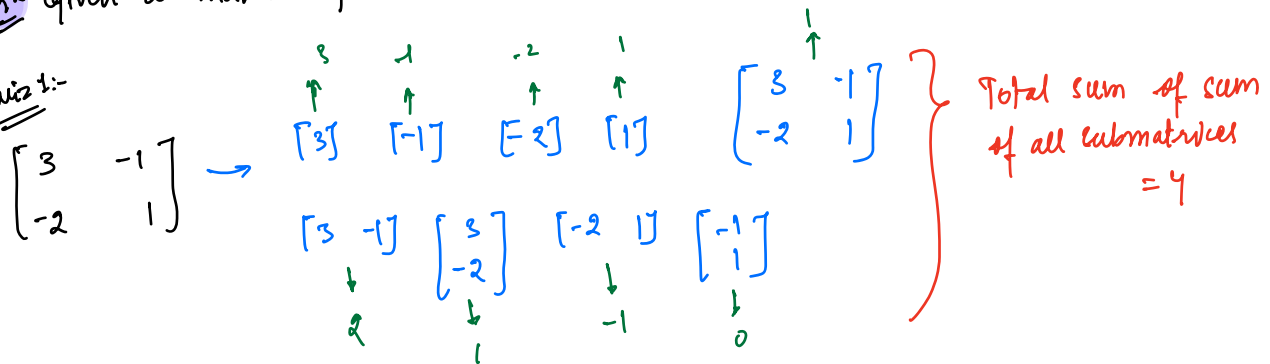
$$T.C = O(N \times M + N \times M + Q \times O(1))$$

$$= O(N \times M + Q)$$

$$S.C = O(1)$$

Q2 Given a matrix of size $N \times M$. Calculate sum of all submatrices sum.

Quiz 1:-

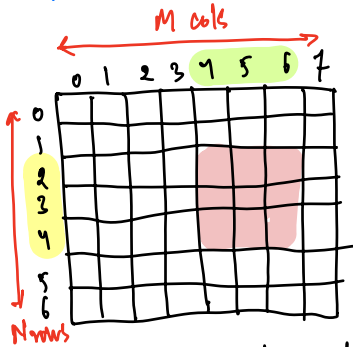


$$3 \times 4 + (-1) \times 4 + (-2) \times 4 + 1 \times 4 \Rightarrow 12 - 4 - 8 + 4 \Rightarrow 4$$

Solution 1:- Generate all submatrices and add sum.

$N \times M$ matrix

submatrices =



subarrays of

- rows: $\frac{N(N+1)}{2}$
- cols: $\frac{M(M+1)}{2}$

$$\# \text{ submatrices} = \frac{N(N+1) \times M(M+1)}{4}$$

Using pf sum matrix Approach.

$$T.C = O(N \times M + Q)$$

$\rightarrow$ # of submatrices.

$$= O(N \times M + N^2 M^2)$$

$$T.C = O(N^2 M^2)$$

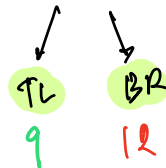
$\rightarrow$ Optimize. $\rightarrow$ Contribution technique.

$$\text{Sum of all submatrices sum} = \left[\begin{array}{c} \text{No. of times mat[i][j]} \\ \text{comes in all submatrices} \end{array} \right] \times \text{mat[i][j]}$$

	0	1	2	3	4	5
0	✓	✓	✓			
1	✓	✓	✓			
2	✓	✓	✓	✓	✓	✓
3			✓	✓	✓	✓
4			✓	✓	✓	✓

→ In how many submatrices this element at (2,2) is present.

How to fix a submatrix



total possibilities = $9 * 12 = 108$ submatrices.

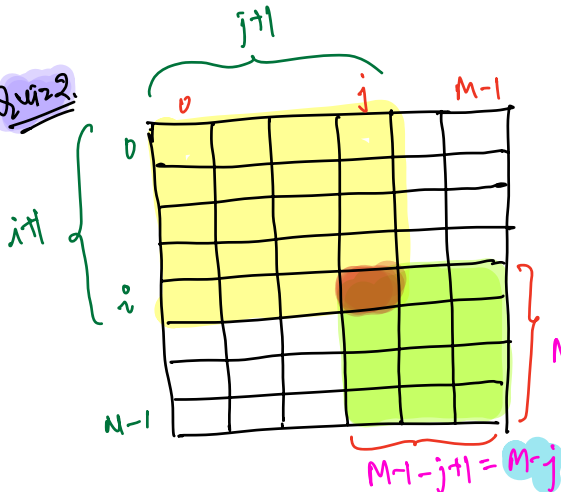
Quiz 1 Given a matrix of size $4 * 5$. no. of submatrices contain mat[1][2]

	0	1	2	3	4
0	✓	✓	✓		
1	✓	✓	✓	✓	✓
2			✓	✓	✓
3			✓	✓	✓

TL = 6
BR = 9

total submatrices which contain mat[1][2] = $6 * 9 = 54$

Quiz 2



TL options = $(i+1) * (j+1)$
BR options = $(N-i) * (M-j)$

$N-1-i+1 = N-i$

$M-1-j+1 = M-j$

total no. of submatrices which has cell (i, j) in a matrix of size $N * M$
 = TL options * BR options
 = $(i+1) * (j+1) * (N-i) * (M-j)$

Total contribution of $\text{mat}[i][j] = (i+1) * (j+1) * (N-i) * (M-j) * \text{mat}[i][j]$

Pseudo Code:-

$\text{ans} = 0;$

for ($i=0; i < N; i++$) {

for ($j=0; j < M; j++$) {

$TL = (i+1)(j+1);$

$BR = (N-i)(M-j);$

$\text{total} = TL * BR;$

$\text{contri} = \text{total} * \text{mat}[i][j];$

$\text{ans} = \text{ans} + (\text{contri});$

}

}

$T.C = O(N * M)$

$S.C = O(1)$

Q3 Given a matrix every row is sorted and every column is sorted.

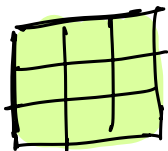
find max submatrix sum.

e.g.

	0	1	2	3
0	-20	-16	-4	8
1	-10	-8	12	14
2	-1	6	21	30
3	5	7	28	72

1) Find all submatrices and get max sum.
 $T.C = O(N^2 M^2)$; $S.C = O(NM)$

e.g. 1. All elements are +ve



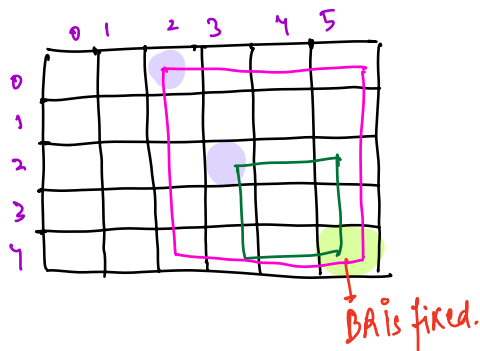
e.g. 2. All elements are -ve

-50	-40	-30	-20
-41	-20	-23	-18
-35	-25	-19	-16
-29	-20	-18	-14

Quiz 4 Given matrix of $N \times M$ with above conditions.
 coordinate for largest value?

mat[N-1][M-1]

→ Ans submatrix sum should always have this-



Ans submatrix

↙
 GL
 Consider everything.

↘
 BR
 (fixed)

ans = INT_MIN or mat[0][0]

```
for (i=0; i < N; i++) {
```

```
    for (j=0; j < M; j++) {
```

```
        sum = (i, j); BR = (N-1, M-1)
```

// You can get the sum of your submatrix using your pf[1][1].

// Refer Quest.

```
        ans = max(ans, sum);
```

```
    }
```

```
}
```

```
return ans;
```

for loop
 $T.C = O(N \times M + N \times M)$

pf[1][1]

$= O(N \times M)$

S.C $\rightarrow O(1)$
 $\rightarrow O(N \times M)$

	0	1	2	3
0	-20	-16	-4	8
1	-10	-8	12	14
2	-1	6	21	30
3	5	7	28	42

